Semi-Active Convolutional Neural Networks for Hyperspectral Image Classification

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***Abstract*— Owing to the powerful data representation ability of deep learning (DL) techniques, tremendous progress has been recently made in hyperspectral image (HSI) classification. Convolutional neural network (CNN), as a main part of the DL family, has been proven to be considerably effective to extract spatial–spectral features for HSIs. Nevertheless, its classification performance, to a great extent, depends on the quality and quantity of samples in the network training process. To select those samples, either labeled or unlabeled, which can be used to enhance the generalization ability of CNNs and further improve the classification accuracy, we propose an iterative semi-supervised CNNs framework by means of active learning and superpixel segmentation techniques, dubbed as semi-active CNNs (SA-CNNs), for HSI classification. More specifically, we start to pretrain a CNN-based model on a small-scale unbiased labeled set and infer unlabeled data using the trained model, i.e., generating pseudolabels. Then, the reliable samples, which consist of two parts: high label homogeneity and most informativeness, are actively selected from superpixel segments. These selected labeled and unlabeled samples with their labels and pseudolabels are refed into the next-round network training. Moreover, three different schedules, i.e., *log*-, *exp*-, and *linear*-schedules, are progressively adopted to fully explore their potentials in sample selection, until a labeling budget is finally reached. Extensive experiments are conducted on three benchmark HSI datasets, demonstrating substantial performance improvements of the proposed SA-CNNs over other similar competitors.**

***Index Terms*— Active learning, classification, convolutional neural network (CNN), deep learning (DL), hyperspectral, pseudolabel, remote sensing, semi-supervised learning, superpixel segmentation.**

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NOMENCLATURES AND NOTATIONS

AL	Active learning.
BN	Batch normalization.
BvSB	Best versus second best.
CCF	Canonical correlation forest.
CNN	Convolutional neural network.
CRNN	Convolutional recurrent neural network.
DL	Deep learning.
FC	Fully connected.
GSD	Ground sampling distance.
HS(I)	Hyperspectral (image).
KNN	K nearest neighbor.
MLR	Multinomial logistic regression.
MRF	Markov random field.
PCA	Principal component analysis.
ReLU	Rectified linear unit.
RNN	Recurrent neural networks.
RS	Remote sensing.
SLIC	Simple linear iterative clustering.
SSAL	Semi-supervised active learning.
SSL	Semi-supervised learning.
SVM	Support vector machine.
x, y	D-dimensional spectral signature and category annotation of an HSI pixel (or patch).
$\mathcal{D}_{\text{train(test)}}$	Training (Testing) set.
$\mathcal{D}_{L_0}$	Initial labeled set.
$\mathcal{D}_U$	Unlabeled set.
$\mathcal{D}_A$	Auxiliary set selected by AL.
$\mathcal{D}_{\text{PL}}$	Pseudolabeled set.
M	Budget, i.e., total number of samples that can be selected.
$\mathcal{T}_*^{(t)}$	Schedule at t-th training iteration, where $*$ could be *log*, *exp*, and *linear*.

I. INTRODUCTION

THE rapid development recently achieved in the RS technology has enabled users with tremendous airborne and

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spaceborne imaging products. Among them, the HS imagery collected by HS instruments, such as Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), Reflective Optics System Imaging Spectrometer (RsOSIS), and ITRES CASI-1500, has been gaining a surge of interest by researchers in the RS community, such as image reconstruction [1], [2], [3], land cover land use classification [4], [5], [6], [7], spectral unmixing [8], [9], [10], [11], target detection [12], [13], [14], [15], and image fusion and super-resolution [16], [17], [18], [19]. The detailed spectral information contained in their hundreds of continuous and narrow spectral wavelengths largely boosts the potential on a multitude of applications, particularly facilitating a long-standing and active topic, namely, hyperspectral image (HSI) classification, which is targeted to distinguish materials of interest based on spectrum contents. Albeit substantiated to be more discriminative with much higher spectral resolution than the multispectral ones, the HSIs, hampered by spectral variability [20] and the curse of dimensionality (also known as the Hughes phenomenon) [21], are still in need of careful processing in practice.

The initial strategies proposed to alleviate high-dimensionality issues are to directly adopt linear transformations from the spectral view, such as PCA, linear discriminant analysis [22], linearized manifold learning [23], and linear regression [24]. Later, the spatial correlation information is further added into consideration, thereby useful spatial–spectral features can be extracted, yielding favorable classification performance gains comparatively. Some typical works along this line include manifold learning methods [25], low-rank representation methods [26], and profile-based methods [27]. Still, the practicality of those elaborately handcrafted feature extraction algorithms remains limited with sacrifice of capacity and flexibility in learning data-dependent features [28].

More recently, the deep learning (DL) technique has provided a revolutionized tool for realizing end-to-end feature extraction and classification of HSIs [29], [30]. Early stage works start from stacked autoencoders and deep belief networks that are based on the FC deep architectures, and then evolved to convolutional neural networks (CNNs) [28] and RNNs [31], with their advantageous abilities of local perception and sequential dependencies memorizing in the spatial and spectral domains, respectively. Very recently, the transformer-based deep network has shown its great potential in HSI feature extraction and classification [32]. There also have been several efforts made to further ameliorate the classification performance by leveraging multimodal features using CNNs [33] and generative adversarial networks as building blocks [34], [35].

The success of the DL-based methods on HSI classification depends crucially on both the quantity and quality of labeled training data, which, however, can hardly be satisfied in the RS scenarios. With the relatively lower spatial resolution on a large-scale covering area, the HS imagery provided by the spaceborne spectroscopy imaging satellite is prone to suffer from an expensive and time-consuming label annotation issue but meanwhile enjoys a plentiful supply of unlabeled data during classification. Therefore, how to select samples in the investigated scene, either labeled or unlabeled, and make full use of them to enlarge the capability of deep classifiers are becoming more nonnegligible.

To better exploit the supervision information within limited labeled data, a considerable number of approaches attempt to smartly construct the desired training set by means of AL, i.e., automatically finding which samples to annotate that can help classifier training the most. This idea has been well-explained in the extensive literature and demonstrated to be effective when applied to large-scale datasets with complex feature spaces [36]. Specifically, by actively sampling informative pixels according to certain criteria measuring the uncertainty based on the current model, advanced HSI classification algorithms that incorporate AL are always capable of alleviating the need for sufficient labeling and achieving superior performance than classic ones.

On the other hand, inspired by the methodology of SSL, a growing interest has also been paid to address the importance of unlabeled data for HSI classification. In contrast to the AL-based methods under the supervised learning framework, the SSL-based methods focus on, in general, learning the data structure information from both the labeled and unlabeled proportions. By doing so, the SSL-based methods have been widely shown to approach or surpass the classification performance of purely supervised ones, even when only a less number of labeled samples are available.

Bearing the concepts of AL and SSL in mind, their combination can be regarded as a new path to tackle the issue of learning with limited labelings. Excellent researches have been recently proposed by integrating these two complementary techniques in the context of HSI classification and achieved promising performance, as will be shown in Section II. Yet it is noteworthy that, on one hand, their mutual impacts on deep classifiers are still rarely investigated. On the other hand, most of the existing methods rely on complicated designs of feature extraction and selecting criterion, which may limit their practicality and extendibility in wider situations. Therefore, in this article, we put forth a unified SSAL framework by learning CNNs in a semi-active manner. The proposed framework aims at addressing the issue of training samples' insufficiency and improper utilization of unlabeled samples simultaneously from the following two perspectives. First, unlike the existing superpixel-guided deep methods that use it for efficient feature extraction, we tentatively integrate it into the proposed active selection of pseudosupervision by investigating the label consistency within superpixels, as Fig. 1 shows. Furthermore, we readily design AL schedules for iterative and progressive selection of the most informative samples in each round, by which two proposed procedures are expected to promote each other through an interleaving fashion, thus yielding a boosted performance peak that can be attained by the investigated CNN-based HSI classifier.

Three main contributions of the proposed method can be summarized as follows.

1) We devise a novel semi-active HSI classification framework by learning semi-supervised CNNs, SA-CNNs for short, via actively selecting both the labeled and

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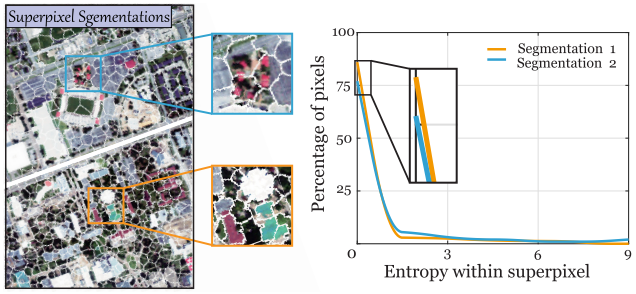


Fig. 1. Statistics of label inconsistency under different superpixel number segmentations using the label entropy within each superpixel. The central superpixels in the middle upper blue box pose relatively higher label inconsistency than those in the lower orange box, while both the segmentations can generate large proportions of superpixels with consistent labels, corresponding to low-entropy side beneath 1 in the right plot.

unlabeled data in an iterative fashion, allowing CNNs to learn more discriminative features using limited labeled data.

2) We show that three intuitive but effective schedules for AL can well fit the network training by progressively adding the actively selected training samples, further enabling the trained deep classifier with more data-adaptive ability to generalize.
3) We conduct extensive comparison experiments on three benchmark HSI classification datasets to show at least 2% accuracy improvements of the proposed SA-CNNs with acceptable computational cost and comprehensive ablation studies to verify its rationality and effectiveness.

The remainder of this article is organized as follows. A brief review of related works is introduced in Section II. Section III proposes our SA-CNNs model and effective schedules for AL. In Section IV, we present the elaborately designed experiments, results, and discussions. Conclusions and future prospects are finally given in Section V.

II. Related Work

In this section, we briefly review typically related works incorporating the strategies of AL, SSL, and SSAL, with either shallow or deep architectures, for the task of HSI classification.

A. HSI Classification With AL

The AL strategy facilitates HSI classifiers to reduce their reliance on a large labeled set by interactively selecting informative samples that are most beneficial for enhancing the classification performance. Extensive families of sampling criteria have been proposed into conventional feature extraction algorithms in the past decades, mainly including the *committee*-based, the *uncertainty*-based, and the *posterior probability*-based heuristics. Among them, the latter two families are more popular since they are more computationally efficient than the committee-based scheme, which typically requires training a committee of classifiers using different hypotheses [37]. To name a few, Li *et al.* first introduced an entropy-based AL strategy into the MLR model [38], and based on posteriors, the breaking ties algorithm and its modified version were used under the Bayesian probabilistic framework in [39]. Di and Crawford then thoroughly explored HSI view generation methods to construct the ensembles in the context of AL [40]. Another widely used criterion is the multiclass-level uncertainty, which can effectively address the multiclass problem using a one-against-all multiclass architecture of binary SVM [41]. More recently proposed DL-based methods that incorporate AL typically include early attempts using restricted Boltzmann machine [42], Bayesian CNNs combining multiple criteria in [43], and the state-of-the-art (SOTA) CNNs postprocessed by MRF [44].

B. HSI Classification With SSL

The SSL framework has received considerable attention these years with the ever-growing availability of unlabeled samples in newly acquired HSI datasets. Unlike the aforementioned AL strategy focusing on labeled proportion, the goal of SSL is to enhance the generalization ability of the classifier by jointly leveraging labeled and unlabeled data. Advanced SSL methods (e.g., FixMatch [45]) have been recently proposed by adopting consistency regularization on perturbed versions of same unlabeled data. To differ from the classification application of natural images, in the RS scenarios, one of the earliest ideas, *self-learning*, was successfully introduced for semi-supervised HSI classification by repeatedly training the probabilistic classifier with a subset of predicted labels, in which the spatial information is considered during the candidate pool inference [46]. Wu and Prasad proposed a clustering constrained SSL model using pseudolabels produced by CRNNs [47]. Graph information is another type of useful guidance, and Hong *et al.* incorporated it into an iterative multitask regression framework [48], thus refining the pseudolabels progressively. They further extended the regression framework by aligning pixels and superpixels, yielding a spatial–spectral dimensionality reduction in HSIs [49].

C. HSI Classification With SSAL

Since both the AL and SSL paradigms share the benefit of alleviating need for sufficient labeling, it is natural to consider their combination to simultaneously exploit labeled and unlabeled information. At first, AL strategies were commonly adopted to facilitate SSL. In [38], an efficient AL-integrated SSL algorithm was first implemented to learn the posterior class distributions and then followed by segmentation based on a multilevel logistic prior. Muñoz-Marí *et al.* proposed to build a hierarchical clustering tree with the help of the active query strategy for unlabeled samples with high uncertainty [50]. Besides, Zhang *et al.* [51] ably introduced the updated spatial information by feature and sample augmentation in an iterative fashion. The superpixel-based similarity assumption was first used along with breaking ties sampling criterion in [52], to assist sample selection and pseudolabeling. However, the aforementioned methods are all based on the conventional classifiers such as hierarchical tree, MLR, and SVM. Despite their satisfying performance, a more general and practical DL-based framework that can fully exploit the potential of both the labeled and unlabeled data remains to be studied.

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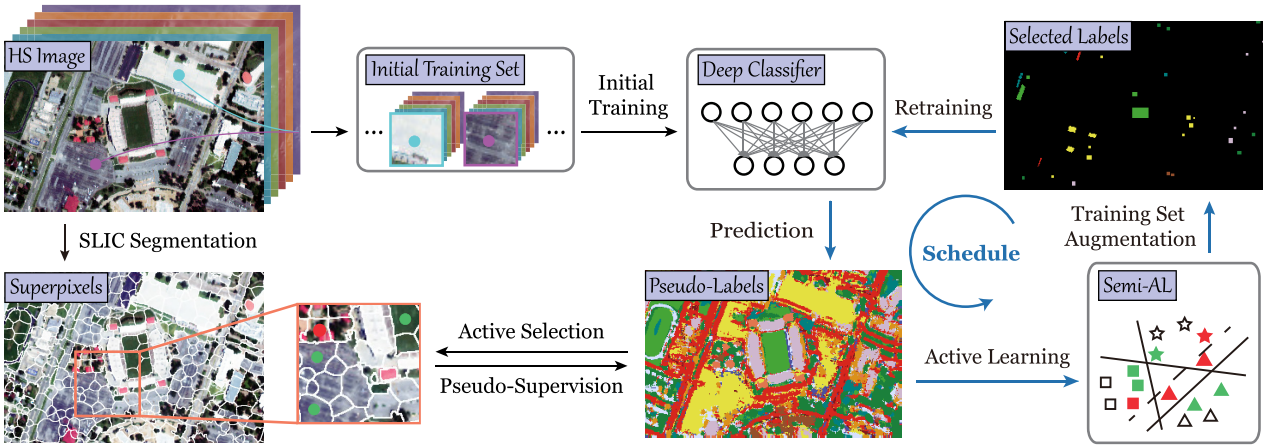


Fig. 2. Illustration of the proposed SA-CNNs' framework for HSI classification. Our network is trained in a progressive manner by simultaneous exploitation of supervised and pseudosupervised information. In each iteration, we first actively select those samples that generate high label consistency based on SLIC, like green marked superpixels, with their pseudolabels to construct a reliable pseudolabeled set. We then conduct AL procedure under three proposed schedules for the remaining samples that pose highly inconsistent semantics in red doted superpixel. Through this manner, our framework is expected to fully exploit the knowledge of the unlabeled part and further boost the efficiency and effectiveness of AL and achieve semi-AL.

III. Proposed Method

In this section, we first give a brief overview of the proposed SA-CNNs, to show the differences between our method and the previous ones. Then, we detail the main body of SA-CNNs with necessary preliminaries. Finally, three effective AL schedules are presented, which enable our networks to be trained with data-adaptive generalization ability.

A. Method Overview

To further eliminate the strong reliance of deep models on a large number of labeled training samples, we propose the semi-active learning framework, i.e., simultaneous active selection of both the labeled and unlabeled data, by which more global data structure information can be exploited and the interrelated supervision and *pseudosupervision* allow facilitating each other in a progressive manner, as the workflow in Fig. 2 illustrates. Moreover, we integrate the proposed SA-CNNs with three AL schedules designed against data distribution discrepancies between the train and test sets, and thus empower our method to generalize preferably in dealing with various kinds of HSI datasets.

B. Superpixel Segmentation

Superpixel segmentation has been embraced by the RS community with an enthusiastic interest during the past decades, especially for the task of HSI classification. It was used by means of probabilistic fusion with pixel-level estimation by Li *et al.* in [53], and further integrated with decision-level fusion [54]. It captures low-level image properties and groups neighboring pixels into irregularly shaped superpixels [55]. Representative superpixel segmentation algorithms include SLIC [55] and superpixel sampling network incorporating differentiable SLIC [56]. Different from previous methods focusing on the potential of superpixels in providing an efficient feature representation, we note that the high label consistency within superpixels, as Fig. 1 depicts, can be exploited to enable the proposed semi-active learning strategy. It is worth noting that such oversegmentations with varying parameters behave relatively consistent tendency in terms of label consistency and also are free from evidently additional computation burden.

C. Proposed SA-CNNs

Let $\mathcal{D} = \{\mathbf{X}, \mathbf{y}\}$ denote the investigated HSI dataset, where $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\} \in \mathbb{R}^{N \times D}$ collects D-dimensional spectral signatures of N pixels of the interest, and $\mathbf{y} \in \mathbb{R}^N$ is the label vector, each element of which represents the land-use or land-cover category for each pixel, given a label set $\mathcal{C} = \{c_1, c_2, \ldots, c_K\}$ with K predefined classes. Our goal of HSI classification is to adopt a CNN-based deep classifier trained on training set $\mathcal{D}_{\text{train}} = \{\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}\}$ to predict labels $\hat{\mathbf{y}}$ for the remaining test set $\mathcal{D}_{test}$.

Usually, CNNs require $\mathcal{D}_{\text{train}}$ with a sufficiently large quantity and clean labels to attain a promising classification performance. The previous AL-based methods commonly initialize a labeled set $\mathcal{D}_{L_0}$ with $|\mathcal{D}_{L_0}| \ll |\mathcal{D}_{\text{train}}|$, where $|\cdot|$ is the cardinality of a set, and then actively select an auxiliary set with their queried labels $\mathcal{D}_A$, leading an evident classifier performance improvement trained on $\mathcal{D}_{L_0} \cup \mathcal{D}_A$ than $\mathcal{D}_{\text{train}}$ ($|\mathcal{D}_{L_0} \cup \mathcal{D}_A| \leq |\mathcal{D}_{\text{train}}|$), where $\cup$ denotes the union of two sets. However, the information of unlabeled proportion is often overlooked, which in fact plays a dominant role in the high-dimensional feature space. Thus, we show that by proper active selection of both the labeled and unlabeled data, their labels and pseudolabels are able to jointly contribute to learning a semi-supervised CNN for better HSI classification.

We adopt vanilla CNNs with the backbone structure as shown in Fig. 3, without loss of generality, which consists of two convolutional layers along with a max-pooling and a BN layer, two FC layers with an ReLU activation layer after the first one, and a Softmax output layer. In each iteration, the categorical cross-entropy loss used for network training is

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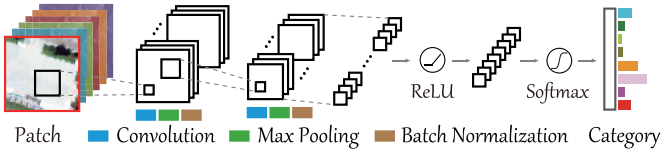


Fig. 3. Backbone network structure of our SA-CNNs for HSI classification. The input HSI patch is sequentially processed by two convolution blocks and two FC layers. The convolution blocks consist of the same structure by cascading a channelwise convolution layer, a max pooling layer, and a BN layer. The FC layers are followed by an ReLU layer and a Softmax layer.

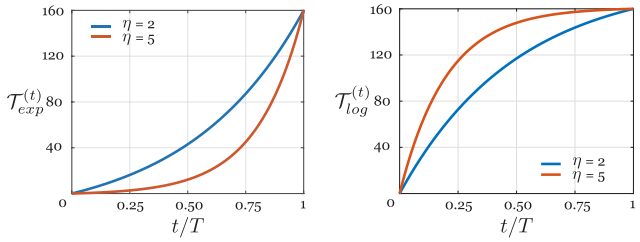


Fig. 4. Illustration of the proposed two AL schedules with an example on the Indian Pines dataset. (a) *exp*-schedule. (b) *log*-schedule.

defined as

$$L(\mathbf{y}, \hat{\mathbf{y}}) = -\sum_{i}\sum_{k=1}^{K} \mathbb{1}_{c_k}(y_i) \log Pr(\hat{y}_i = c_k | \mathbf{x}_i, \Theta) \quad (1)$$

where Θ collects the parameters of CNNs trained on $\mathcal{D}_{\text{train}}$, $\mathbb{1}_{c_k}(\cdot)$ denotes the indicator function, and $Pr(\cdot)$ represents the output score of CNNs for each class, and the condition on $(\mathbf{x}_i, \Theta)$ will be omitted hereafter for notation simplicity.

As usual, we initialize $\mathcal{D}_{\text{train}}$ with a small labeled set $\mathcal{D}_{L_0}$, using unbiased sampling, e.g., 20 for each class of the Houston2013 dataset in our case. The conventional AL-based methods then directly implement an active selection of unlabeled samples and query for human annotations. Among these methods, Cao *et al.* [44] have verified the superiority of the BvSB-based criteria [57] against entropy-based criteria implemented by means of 3D-CNNs, in which the BvSB measuring the margin-based uncertainty in the context of multiclass classification can be defined as

$$B(\hat{y}_i) = Pr(\hat{y}_i = c_B) - Pr(\hat{y}_i = c_{\text{SB}}) \quad (2)$$

where c_B and c_{SB} denote the class indexes that unlabeled sample $\mathbf{x}_i \in \mathcal{D} \setminus \mathcal{D}_{\text{train}}$ belongs to the largest and second largest probabilities, respectively.

Nevertheless, a commonly neglected issue lies in that the limited supervision provided in early iterations inclines to generate an overfitted model that fails to well account for the unlabeled set, further misleading a directly followed AL process. This issue has also been verified in our practice by the fact that the performance is typically not guaranteed with constant improvements when gradually feeding CNNs with actively selected labeled data. Therefore, we are motivated to design a robust AL framework for DL-based HSI classification, which is expected to fulfill the powerful feature extraction potential of CNNs.

In this sense, SSL can serve as an alternative for its ability in leveraging pseudosupervision from unlabeled data, with quite a few SOTA methods showing its effectiveness, either based on conventional MLR model integrated with AL [52] or using elaborately designed CRNNs aided by the Dirichlet process mixture model [47]. However, unlike these methods, we propose to iteratively select both the labeled and unlabeled data, by which pseudosupervision is largely exploited to facilitate conventional AL in a unified framework.

Our intuitive yet clear motivation lies in that the category information within superpixels is highly consistent—as illustrated on an investigated HSI scene in Fig. 1—even under segmentations with varying numbers of superpixels. Based on the current network outputs, we can thus introduce the label entropy to measure the label consistency in each superpixel $\mathcal{S}$

$$E(\hat{\mathbf{y}}_{\mathcal{S}}) = -\sum_{k=1}^{K} P(\hat{y}_i = c_k) \log P(\hat{y}_i = c_k) \quad \forall i \in \mathcal{S} \quad (3)$$

where $P(\cdot)$ denotes the class membership probability. Note that for each superpixel, a high label entropy tends to indicate two cases. The first corresponds to an undersegmented case, i.e., the superpixel groups' pixels with inconsistent semantics, which could possibly be due to an overlarge number of superpixel sets or unidentified pixel feature discrepancies. In the second case, the current classifier inclines to show a lack of confidence in predictions and thus generates probabilities without evident gaps across categories.

Bearing the above discussions in mind, we propose to iteratively perform the following three steps to achieve the semi-supervised learning of our SA-CNNs. First, we set a threshold on $E(\hat{\mathbf{y}}_{\mathcal{S}})$ to divide the unlabeled samples $\mathbf{X}_U$ into two parts. For the part $\mathbf{X}_{\tilde{U}}$ whose label entropy is less than this threshold, we treat them as the reliable samples regarding current classifier and collect their predictions together into the pseudolabeled set $\mathcal{D}_{\text{PL}} = \{\mathbf{X}_{\tilde{U}}, \hat{\mathbf{y}}_{\tilde{U}}\}$. We next use the AL process for the rest of the part samples $\mathbf{X}_U \setminus \mathbf{X}_{\tilde{U}}$, i.e., actively selecting the most confusable candidates $\mathbf{X}_A$ by means of on-the-shelf criteria, such as BvSB, and querying for their annotations $\mathbf{y}_A$ to obtain an augmented labeled set by adding to the initial one $\mathcal{D}_L = \mathcal{D}_{L_0} \cup \{\mathbf{X}_A, \mathbf{y}_A\}$. Finally, the deep classifier can be updated through supervised training on $\mathcal{D}_{\text{train}} = \mathcal{D}_L \cup \mathcal{D}_{\text{PL}}$ and thus expected to ameliorate the classification performance by complementary exploitation of both reliable pseudosupervision and informative supervised information that facilitate each other.

D. AL Schedule

Given the fact that only limited labeled samples are initialized, the feature subspace they expand is likely too vulnerable, for the CNNs, to learn a reliable representation of the data and thereby degrade the following generation of pseudolabels in early stages. Generally, we propose to renew the commonly adopted *linear*-schedule for active selection of labeled samples in a gradually released fashion, rather than picking up them once. In specific, given the *budget* M, i.e., the total number of samples that can be selected, the cumulative adding number

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Algorithm 1 Semi-Supervised CNNs for Semi-Active HSI Classification

Input: HS data $\mathbf{X}$.
Output: Whole image pixel label $\mathbf{y}$.
Initialize Superpixel map $\mathbf{S}$, budget M, schedule $\mathcal{T} \in \{\mathcal{T}_{linear}, \mathcal{T}_{log}, \mathcal{T}_{exp}\}$, initial labeled set $\mathcal{D}_L = \{\mathbf{X}_L, \mathbf{y}_L\}$, AL augmented set $\mathcal{D}_A = \{\}$, pseudolabeled set $\mathcal{D}_{PL} = \{\}$, iteration number $t = 1$ and its maximum T.
while $t < T$ **do**
  Update CNNs' parameters by training on $\mathcal{D}_{train} = \mathcal{D}_L \cup \mathcal{D}_{PL}$;
  Obtain pseudolabels with class probability on $\mathbf{X}_U = \mathbf{X} \setminus \mathbf{X}_L$ to obtain $\hat{\mathbf{y}}_U$;
  Actively select $\mathbf{X}_{\tilde{U}}$ based on $E(\mathbf{y}_L \cup \hat{\mathbf{y}}_U)$ to construct $\mathcal{D}_{PL} = \{\mathbf{X}_{\tilde{U}}, \hat{\mathbf{y}}_{\tilde{U}}\}$;
  Actively select $\mathbf{X}_A^{(t)}$ with size of $\mathcal{T}^{(t)} - \mathcal{T}^{(t-1)}$ from $\mathbf{X}_U \setminus \mathbf{X}_{U_1}$, and query for annotations $\mathbf{y}_A^{(t)}$;
  $\mathcal{D}_A \leftarrow \mathcal{D}_A \cup \{\mathbf{X}_A^{(t)}, \mathbf{y}_A^{(t)}\}$, $\mathcal{D}_L \leftarrow \mathcal{D}_L \cup \mathcal{D}_A$;
  $t \leftarrow t + 1$;
end

of newly labeled samples $\mathcal{T}^{(t)}$ at certain training iteration t follows

$$\mathcal{T}_{\text{linear}}^{(t)} = \frac{t}{T-1} \cdot M \tag{4}$$

where T denotes the maximum iteration number. Note that this schedule serves as an unbiased option when compared with the following two schedules.

Nevertheless, in real scenarios, the users are likely confronted with data distribution discrepancies between the initial training set and the testing set. For example, if the initially sampled data are evenly distributed within the whole feature space, an intuitive conclusion can be made that the current CNNs are prone to learn robust feature representation and thus be able to judge which samples the classifier probably needs the most in the next iteration. Theoretically, we can feed the CNNs with the newly labeled samples as carefully as we can—in extreme cases one sample for each iteration—to reach the full potential of AL, which also requires endless training of the networks. Aiming to attain a trade-off between the performance gain and unbearable computational cost, we consider adding samples according to an *exp*-schedule as follows:

$$\mathcal{T}_{\text{exp}}^{(t)} = \exp\left(\left(\frac{t}{T-1} - 1\right) \cdot \eta\right) \cdot M \tag{5}$$

where η is the scaling parameter controlling the varying speed of the schedule; as Fig. 4 indicates, the bigger the η the steeper the schedule. The basic motivation lies in that the samples added in early stage affect the model more crucially than the later selected ones.

On the contrary, if samples are sparsely distributed, or say there exists evident gap between data distributions, the initially learned CNNs using only a limited training set can hardly output reliable class probabilities for unlabeled samples. In this case, an elaborate selection at the very beginning may jeopardize the model to underfitting the whole dataset due to overfitting the limited amount of initial set. Instead, we propose to adopt the *log*-schedule for SA-CNNs

$$\mathcal{T}_{\text{log}}^{(t)} = \left(1 - \exp\left(-\frac{t}{T-1} \cdot \eta\right)\right) \cdot M \tag{6}$$

to absorb more labeled data at the beginning of iterations and promote a more sound track of subsequent learning.

The entire procedures of our SA-CNNs can be summed up in Algorithm 1 by integrating the proposed AL schedules.

IV. Experiments and Discussion

In this section, we first present description of our investigated datasets for HSI classification. Then we introduce the implementation details and comparison methods to verify the effectiveness of the propose method. Extensive ablation study as well as comparison results and discussions are given at last.

A. Data Description

Three benchmark HSI datasets are selected to validate the HSI classification performance of ours and compared methods.

1) Indian Pines Dataset: The first HSI dataset was gathered by the AVIRIS sensor in June 1992, covering the Indian Pines test site in North-Western Indiana, USA.[1] A subscene consisting of 145 × 145 pixels with a 20-m GSD and 220 spectral bands ranging from 400 to 2500 nm wavelength is widely used, the sample statistics and visualization of which are demonstrated in Table I and Fig. 5, respectively.

2) Pavia University Dataset: The second HSI dataset was gathered by the ROSIS sensor, covering the well-known Pavia University, Pavia, Italy. The valid scene consists of 610 × 340 pixels with a 1.3-m GSD and 103 bands ranging from 430 to 860 nm. Table I summarizes the sample information categorized into nine land cover classes, while Fig. 6 shows its natural composite image and spatial distribution.

3) Houston2013 Dataset: The last HSI dataset was gathered by the ITRES-CASI 1500 sensor in June 2012, covering the University of Houston campus and its surrounding rural area in Texas, USA. It was initially provided for the 2013 IEEE GRSS data fusion contest.[2] This image is of 349 × 1905 pixels at a spatial resolution of 2.5 m and 144 spectral bands ranging from 380 to 1050 nm after necessary preprocessing. Class information about 15 categories of these data is listed in Table I, and the generated true-color composite image and ground-truth map are shown in Fig. 7.

B. Experimental Setup

1) Evaluation Metrics: Three classic evaluation metrics are adopted to quantitatively assess the classification performance of each model, i.e., overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ). OA measures the probability that an individual pixel in the test set will be correctly classified. AA refers to the average of each accuracy per class. Based

[1] https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html
[2] https://www.grss-ieee.org/technical-committees/image-analysis-and-data-fusion/

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TABLE I
SCENE CATEGORIES OF THREE INVESTIGATED HSI DATASETS, WITH SAMPLE NUMBERS OF INITIAL TRAINING SET AND GROUND TRUTH SHOWN FOR EACH CLASS

No.	Indian Pines			Pavia University			Houston2013		
	Class Name	Train (Init.)	Total	Class Name	Train (Init.)	Total	Class Name	Train (Init.)	Total
1	Corn Notill	20	1434	Asphalt	20	6852	Healthy Grass	20	1251
2	Corn Mintill	20	834	Meadows	20	18686	Stressed Grass	20	1254
3	Corn	20	234	Gravel	20	2207	Synthetic Grass	20	697
4	Grass Pasture	20	497	Trees	20	3436	Tree	20	1244
5	Grass Trees	20	747	Metal Sheets	20	1378	Soil	20	1242
6	Hay Windrowed	20	489	Bare Soil	20	5104	Water	20	325
7	Soybean Notill	20	968	Bitumen	20	1356	Residential	20	1268
8	Soybean Mintill	20	2468	Bricks	20	3878	Commercial	20	1244
9	Soybean Clean	20	614	Shadows	20	1026	Road	20	1252
10	Wheat	20	212	-	-	-	Highway	20	1227
11	Woods	20	1294	-	-	-	Railway	20	1235
12	Buildings Grass Trees Drives	20	380	-	-	-	Parking Lot1	20	1233
13	Stone Steel Towers	20	95	-	-	-	Parking Lot2	20	469
14	Alfalfa	6	54	-	-	-	Tennis Court	20	428
15	Grass Pasture Mowed	6	26	-	-	-	Running Track	20	660
16	Oats	6	20	-	-	-	-	-	-
Total	-	278	10366	-	180	43923	-	300	15029

on the confusion matrix, Kappa index takes imbalance in multiclass classification into account and can therefore picture the performance of the classifier more completely. We repeat five-time sets of experiments with randomly selected training samples and report the averaged classification metrics and running time in seconds. The Kappa value is multiplied by 100 to better show standard deviations, which are summarized in brackets under each averaged metrics in Tables VI–VIII.

2) Comparison Methods: We conduct extensive comparison experiments using ten representative methods by following related, including conventional methods such as KNN classifier, SVM, AL integrated MLR (AL-MLR)[3] [38], CCFs [58],[4] SSAL based on MLR (SSAL) [52],[5] and DL-based methods such as 2D-CNNs, 3D-CNNs [59],[6] vanilla CNNs (v-CNNs) that take our backbone networks, CNNs with BvSB criterion and linear schedule-based AL (AL-CNNs), and CNN-AL-MRF[7] [44]. Note that we make appropriate modifications to CNN-AL-MRF, obtaining its semi-supervised version (AM-CNNs) by iteratively training with MRF-smoothed pseudolabels. Some necessary parameter settings are described below.

1) For the KNN classifier, we empirically find that the best results on three datasets are obtained by setting the number of nearest neighbors to 1.
2) For the SVM, we adopt friendly implementation by the LIBSVM toolbox.[8] In our case, only the fold number of cross validation needs to be manually set to 5.
3) For AL-MLR, the multilevel logistic smoothness parameter is tuned to {3, 6, 3} for three datasets, respectively.
4) For CCFs, the number of decision trees is set to 200.
5) For SSAL, there are four variants, i.e., N, S, SN, and SDP, in the given toolbox, and we follow the parameter settings as [52] tells and select the best model to report results.
6) For 2D-CNNs, we follow the one in [60] and add a preprocessing with PCA using ten reduced dimensions.
7) The spatial size of the input patch in 3D-CNNs is set to 7×7.
8) The smooth factor of MRF in AM-CNNs is set to 10.
9) For our SA-CNNs, we consider the input HSI patch of $8 \times 8 \times D$, the first and second convolution layers of 20 filters with the size of 3×3 and 2×2, respectively, pooling layers of 2×2 kernel and 2 stride, the first FC layer of 500 units, and the second FC layer of size K, which are also same for v-CNNs, AL-CNNs, and AM-CNNs.

3) Implementation Details: To guarantee a fair comparison to the utmost extent, we propose a two-stage procedure to synthetically obtain the final test set. In the first stage, a relatively small labeled set $\mathcal{D}_{L_0}$ is chosen as the initial training set by evenly random sampling across classes from the whole data $\mathcal{D}$. Then, given a certain size of M, or the budget as we call, for the methods without considering AL, we keep random sampling to obtain a labeled set $\mathcal{D}_{L_1}$, whereas $\{\mathcal{D}_{L_q}\}_{q=2}^{Q}$ are the query sets of different AL-based methods conditioned on $\mathcal{D}_{L_0}$, such that $|\mathcal{D}_{L_q}| = M,\ q = 1, \ldots, Q$. Note that all the evaluations are finally proceeded on the remaining set, which is regarded as the unlabeled testing set, i.e., $\mathcal{D}_U := \mathcal{D} - \cup_{q=0}^{Q} \mathcal{D}_{L_q}$. The detailed sizes of $\mathcal{D}_{L_0}$s for three datasets are also summarized in Table I. We follow the experimental setting in [43] by setting 5% of the ground-truth data as the validation set for selection of involved hyperparameters (such as learning rate, epochs, and AL schedule) and the remaining 95% for testing after the initial training set $\mathcal{D}_{L_0}$ is selected. All the following experiments are performed on a Windows System PC with 32-GB memory and an Nvidia GeForce GTX 1080Ti GPU card. Specifically, we implement the proposed SA-CNN

[3]http://www.lx.it.pt/ jun/demos.html
[4]https://github.com/twgr/ccfs
[5]https://github.com/Vikki23/Demo_SSAL_SDP
[6]https://github.com/nshaud/DeepHyperX
[7]https://github.com/xiangyongcao/CNN-AL-MRF
[8]https://www.csie.ntu.edu.tw/ cjlin/libsvm/

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TABLE II

QUANTITATIVE PERFORMANCE COMPARISON OF VARYING NUMBERS OF SUPERPIXELS FOR SA-CNNS WITH A FIXED *Linear*-SCHEDULE ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Metric	Number of Superpixels				
	500	1000	1500	2000	2500
OA (%)	87.22	**88.68**	87.56	86.03	84.37
AA (%)	91.39	**91.23**	90.62	90.31	89.01
κ	85.49	**87.05**	85.89	84.08	82.66

approach using the MatConvNet library, the 3D-CNNs using the PyTorch framework, and the other compared methods using MATLAB R2016b. We initialize the learning rate as 1e-3 and gradually reduce it by multiplying 0.9 after each 20 epochs out of a total 200 epochs per round. The parameter of the weight decay is set to 5e-4. To better illustrate the performance comparison of the investigated strategies, we do not adopt any data augmentation techniques for all the competitors.

C. Ablation Study

We first conduct ablation experiments on the Indian Pines dataset to validate the effectiveness of each proposed technique and also demonstrate the process of achieving final results.

1) Superpixel Number Sensitivity: As above-mentioned in Section III-B., there have been many superpixel segmentation methods proposed recently. We choose the SLIC algorithm [55] for its friendly implementation and stable performance. By default, the only parameter that needs to be manually set is the number of superpixels. With a fixed linear schedule with a 160 budget, five sets of experiments by ranging this parameter evenly from 500 to 2500 are carried out. We summarize the results in Table II and can draw two main conclusions. Overall, the performance of SA-CNNs keeps stable due to the fact that superpixel segmentation always produces an oversegmentation result when the parameter is set moderately. However, an evident decrease in terms of three indices appears when the parameter is set overlarge. This may because negative pseudolabels are more likely to occupy a single superpixel in this situation and thus fake the active selection of reliable pseudosupervision.

2) Effectiveness Verification: We then compare our method with different AL strategies to evaluate its effectiveness. In addition to three commonly adopted criteria, namely, random sampling, entropy, and BvSB, we consider a novel CCF-based one by regarding the probability of decision trees' output as a measure of uncertainty. We show the results under four sets of budgets ranging from 40 to 160 with a linear schedule given in Table III. We can first observe that random strategy behaves best in terms of OA in 40 budget cases. A possible explanation for this observation is that the network overfits the initial training set with limited size and therefore deteriorates the AL process in early stages. When the budget increases to sufficient levels, the AL methods start to offset previous deterioration and bring desirable performance gain. Among them, the results of CCFs can slightly surpass those of entropy, while BvSB is able to provide a satisfying performance only second to ours.

TABLE III

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT AL METHODS WITH A FIXED 1000 SUPERPIXEL NUMBER AND *Linear*-SCHEDULE ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Budget M	Metric	AL Method				
		Random	CCFs	Entropy	BvSB	Ours
40	OA	**74.11**	69.56	69.43	73.84	74.02
	AA	83.09	82.02	82.09	84.85	**85.46**
	κ	69.17	66.98	68.15	70.57	**71.74**
80	OA	74.62	72.59	70.97	74.34	**81.24**
	AA	83.37	83.08	80.58	82.55	**88.63**
	κ	71.70	69.46	68.26	72.91	**78.97**
120	OA	77.85	76.16	75.23	81.41	**86.90**
	AA	85.22	84.88	83.64	86.26	**90.02**
	κ	77.25	76.22	75.93	80.05	**85.83**
160	OA	80.87	80.72	79.53	86.48	**88.68**
	AA	86.08	87.19	85.77	89.84	**91.23**
	κ	79.01	79.22	78.53	83.52	**87.05**

TABLE IV

QUANTITATIVE PERFORMANCE COMPARISON OF FINE-TUNING AND REINITIALIZATION IN EACH ITERATION FOR SA-CNNS WITH A FIXED 1000 SUPERPIXEL NUMBER AND *Linear*-SCHEDULE ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

$\lvert\mathcal{D}_L\rvert$	Fine-Tuning		Re-Initialization	
	$\lvert\mathcal{D}_{PL}\rvert$	Metric	$\lvert\mathcal{D}_{PL}\rvert$	Metric
318	3437	**74.13**	3437	74.02
		85.38		**85.46**
		71.29		**71.74**
358	4665	80.06	4869	**81.24**
		86.78		**88.63**
		76.92		**78.97**
398	5711	85.04	6263	**86.90**
		88.67		**90.02**
		82.98		**85.83**
438	6143	87.07	6740	**88.68**
		90.04		**91.23**
		85.43		**87.05**

3) Fine-Tuning Versus ReInitialization: Note that there exists another kind of performance barrier in detailed implementation besides the possible overfitting risk in the early stages. In Table IV, we report the numbers of both labeled and pseudolabeled samples selected and the corresponding performance metrics in each iteration when adopting different choices of fine-tuning and reinitialization. The classification performance of reinitialization outperforms that of fine-tuning almost in all the cases with more potential of reliable pseudosupervision being automatically excavated. The reason behind this phenomenon may be that the drift of sample distribution will cause errors to accumulate as the iteration progresses. Therefore, we propose to reinitialize our SA-CNNs in each iteration without bringing significant computational overheads compared with the fine-tuning strategy.

4) AL Schedule: We finally study the influence of applying different AL schedules. The experimental results in Table V are obtained under a 1000 superpixel number and 160 budget setting on the Indian Pines dataset. We also include the result of vanilla AL by adding samples at once for better

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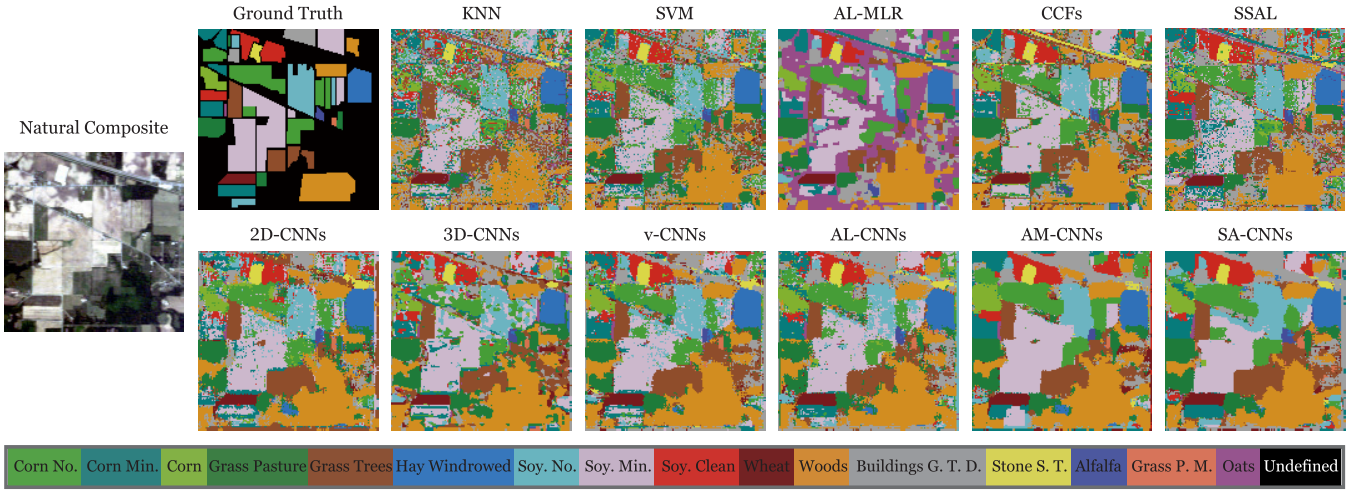


Fig. 5. Illustration of natural composite image, label ground truth, and classification maps obtained by the compared methods on the Indian Pines dataset.

TABLE V

QUANTITATIVE PERFORMANCE COMPARISON OF VARYING SCHEDULES FOR SA-CNNS WITH A FIXED 1000 SUPERPIXEL NUMBER ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Metric	AL Schedule			
	Once	Exp	Linear	Log
OA (%)	83.80	88.21	88.68	**90.04**
AA (%)	88.68	91.62	91.23	**92.99**
κ	81.61	86.76	87.05	**89.16**

comparison. Not surprisingly, our proposed three schedules are able to yield remarkable improvements over the baseline, which shows that progressive strategy is more effective in screening informative samples. Moreover, the best result is attained by applying the log schedule. We conjecture that a large intraclass discrepancy in feature space may occur in this dataset.

D. Comparison Results and Discussion

We further evaluate our method on three benchmark HSI datasets with comparison to other HSI classification methods.

1) Indian Pines Dataset: We consider the following experimental setting of our SA-CNNs, 1000 superpixel number, 160 budget, and log AL schedule. The quantitative results and classification maps of all the competitors and the proposed method are shown in Table VI and Fig. 5, respectively.

From the table, we can see that KNN as a primitive method can only provide a barely acceptable classification result, while SVM evidently outperforms it. Although AL-MLR falls in between the previous two methods, it reaches the best accuracy on class No. 14. CCFs achieve the best result of the conventional methods in terms of three metrics, which also surpasses those of 2D-CNNs and 3D-CNNs, showing its fairly well ability in capturing correlations underlying input HS pixel vectors. For the other CNN-based methods, AL-CNNs and AM-CNNs can largely improve the performance of v-CNNs with 4%–7% owing to their successful application of the AL and MRF techniques, respectively. As expected, our SA-CNNs make a further breakthrough on the basis of AL-CNNs, even with an obvious advantage of nearly 2% in all the metrics compared with SOTA AM-CNNs. Note that SA-CNN is also the method that attains the best results on all the classes, with 100% accuracies on five classes.

A similar tendency can be observed from Fig. 5. In detail, the visual results of KNN and SVM contain less smooth area, while AL-MLR behaves much better at this point even without spatial modeling. CCFs, v-CNNs, and AL-CNNs can outperform previous methods, but still making confusion error in the area of soybean plant. Our SA-CNN method obtains the classification map that is not only the most accurate but also as smooth as that of the MRF-aided AM-CNNs. These results reveal the fact that proper use of complementary labeled and pseudolabeled supervision in our method can benefit the CNN-based classifier to a great extent.

Regarding the running time of the compared methods, we can note that conventional methods such as KNN, SVM, and CCFs behave much faster than the other methods. AL-MLR is second to them and SSAL costs considerably more time on using unlabeled data. 3D-CNNs are much time-consuming than 2D-CNNs and v-CNNs due to their cross-channel convolution style. While our SA-CNN method in-between AL-CNNs and AM-CNNs is capable of achieving a substantial performance gain without paying too prohibitive computational price.

2) Pavia University Dataset: In this dataset, our experimental settings of SA-CNNs are 5000 superpixel numbers, 320 budget, and exp AL schedule. Table VII lists the quantitative results, and the corresponding classification maps of different methods are shown in Fig. 6.

Overall, the classification results of KNN, SVM, and AL-MLR are relatively inferior to other methods, but AL-MLR performs best on three particular classes due to its active and effective utilization of unlabeled data. 2D-CNNs achieve better metrics compared with 3D-CNNs and v-CNNs, which

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TABLE VI

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA, AA, AND κ ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class No.	KNN	SVM	AL-MLR	CCFs	SSAL	2D-CNNs	3D-CNNs	v-CNNs	AL-CNNs	AM-CNNs	SA-CNNs
1	46.71	71.17	60.30	68.31	68.25	73.61	65.52	75.21	84.69	**85.61**	84.95
2	46.58	65.63	62.27	56.54	68.55	44.21	53.18	67.87	83.44	74.65	**96.37**
3	53.77	78.77	88.21	89.15	71.50	87.26	77.83	92.92	95.31	**100.00**	98.59
4	89.08	93.58	87.37	94.22	93.08	82.23	92.72	84.37	87.05	**96.58**	93.62
5	89.79	91.61	81.96	96.22	92.85	97.90	92.59	96.20	96.53	97.37	**100.00**
6	95.16	99.12	98.24	99.12	96.59	99.78	98.24	**100.00**	**100.00**	99.57	**100.00**
7	62.06	61.41	45.02	72.56	67.72	73.10	61.95	77.92	65.42	79.49	**89.80**
8	67.22	73.41	64.77	**87.20**	58.54	71.96	80.31	73.99	83.59	84.00	84.48
9	43.74	71.53	55.75	72.90	86.70	65.35	77.02	69.47	80.07	**86.75**	79.65
10	95.26	98.95	99.47	99.47	98.96	99.47	98.42	**100.00**	**100.00**	**100.00**	**100.00**
11	80.08	90.92	94.26	96.89	95.45	98.25	93.94	97.69	95.48	**98.66**	95.97
12	41.69	60.00	76.90	54.65	72.50	80.00	91.27	89.01	99.16	98.87	**100.00**
13	94.52	97.26	97.26	97.26	90.67	**100.00**	97.26	98.63	94.67	**100.00**	**100.00**
14	68.89	73.33	**82.22**	66.67	70.83	71.11	75.56	66.67	**82.22**	66.67	**82.22**
15	84.21	89.47	84.21	89.47	95.00	94.74	89.47	**100.00**	**100.00**	73.68	**100.00**
16	71.43	**100.00**	**100.00**	**100.00**	**100.00**	78.57	92.86	**100.00**	85.71	**100.00**	92.86
OA (%)	65.87 (0.98)	77.19 (1.76)	71.16 (1.65)	81.71 (1.02)	75.73 (1.21)	78.01 (1.69)	78.95 (1.92)	81.84 (1.57)	86.45 (1.08)	88.75 (1.11)	**90.04** (1.03)
AA (%)	70.64 (1.01)	82.26 (4.05)	79.62 (2.23)	83.78 (1.49)	82.95 (1.24)	84.35 (1.10)	83.64 (1.33)	86.87 (1.21)	89.58 (1.02)	90.12 (1.15)	**92.99** (0.99)
κ	61.28 (1.07)	74.03 (2.36)	68.01 (1.77)	78.97 (1.20)	72.61 (1.18)	74.99 (1.02)	75.91 (1.18)	79.62 (1.06)	84.94 (1.09)	87.48 (1.11)	**89.16** (1.08)
Time (s)	1.11	6.48	140.83	10.92	569.94	743.26	2670.47	944.54	3151.22	3698.69	3443.44

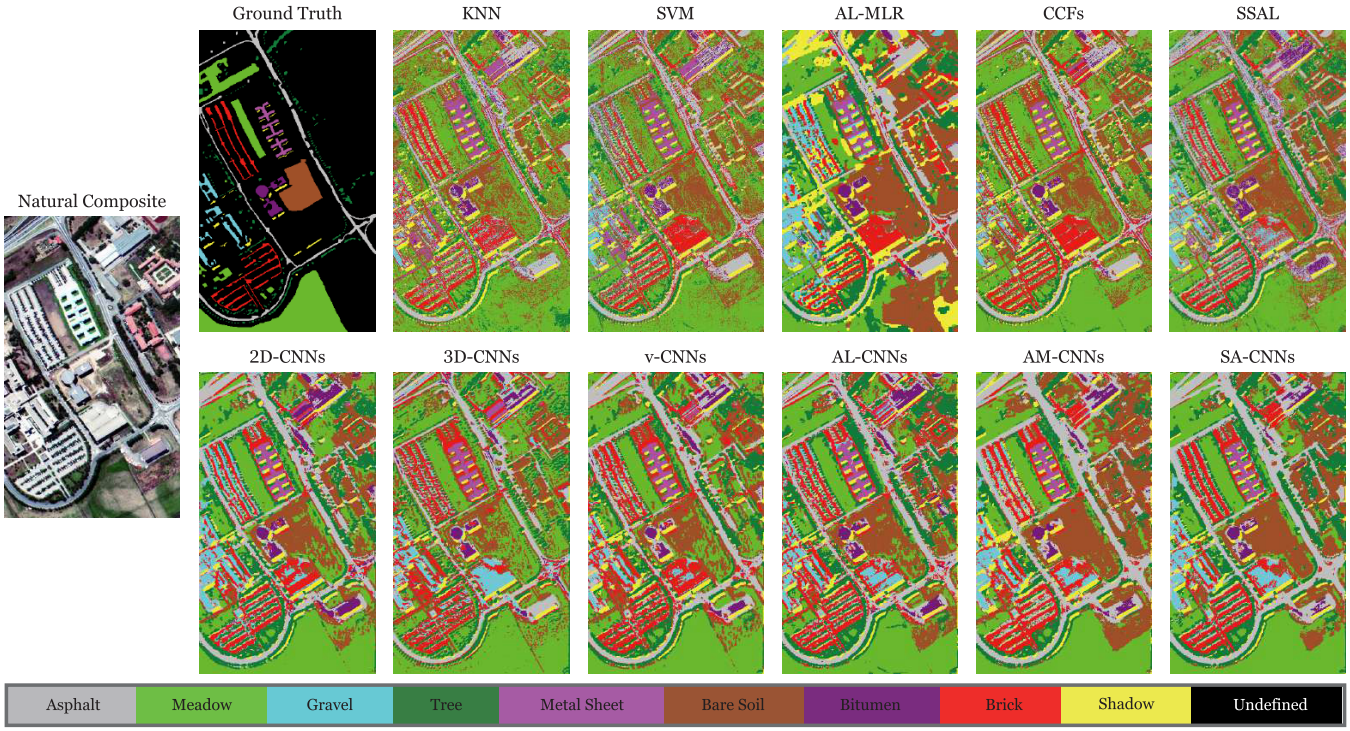


Fig. 6. Illustration of natural composite image, label ground truth, and classification maps obtained by the compared methods on the Pavia University dataset.

indicates the importance of preprocessing along with the spectral domain. Furthermore, AL-CNNs and AM-CNNs exhibit comparable results better than 2D-CNNs, and our SA-CNNs attain a substantial improvement both quantitatively and qualitatively. As shown in Fig. 6, the map of SA-CNNs presents more clear object patterns that are closer to the ground truth.

It is also worth noting that quite a number of advanced methods resort to an MRF-like post-processing procedure based on probabilistic maps. However, in our practice, we found it still suffers from parameter sensitivity issues, which can easily result in a sacrifice of AA due to its overlooking of the class distribution knowledge. On the contrary, the proposed semi-active framework can, on one hand, strengthen

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TABLE VII

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA, AA, AND κ ON THE PAVIA UNIVERSITY DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class No.	KNN	SVM	AL-MLR	CCFs	SSAL	2D-CNNs	3D-CNNs	v-CNNs	AL-CNNs	AM-CNNs	SA-CNNs
1	70.36	78.68	77.42	80.56	74.18	86.71	78.89	92.62	94.03	**99.54**	98.65
2	80.05	83.52	60.02	96.07	90.82	99.01	96.00	98.44	97.84	98.34	**99.95**
3	61.40	69.66	**81.14**	62.46	93.77	74.90	75.68	58.47	68.72	73.05	76.84
4	72.02	79.62	94.42	88.40	92.43	97.33	91.07	96.30	**97.94**	83.21	95.15
5	99.11	99.34	81.76	99.85	97.72	97.05	**100.00**	99.34	99.56	99.93	**100.00**
6	61.87	80.56	94.98	81.37	85.27	89.14	59.99	81.33	90.65	98.50	**99.11**
7	80.11	73.50	**94.14**	86.94	88.31	85.89	79.28	65.99	83.03	72.40	79.60
8	78.34	71.83	72.01	91.37	78.47	84.34	73.83	91.35	89.47	93.73	**97.41**
9	99.60	99.50	**99.80**	99.70	99.40	97.80	96.70	98.50	99.50	98.31	98.91
OA (%)	75.76 (0.84)	81.71 (0.43)	74.23 (1.18)	89.17 (1.24)	86.59 (0.96)	92.82 (1.40)	85.42 (1.29)	91.79 (1.41)	93.88 (1.02)	94.95 (1.11)	**97.50** (0.98)
AA (%)	78.10 (0.53)	82.00 (3.45)	83.97 (1.01)	87.41 (0.41)	87.82 (0.98)	90.24 (1.44)	83.49 (1.97)	86.92 (1.28)	91.19 (0.88)	90.32 (1.12)	**94.29** (0.90)
κ	68.57 (0.98)	76.26 (0.55)	69.03 (0.97)	85.72 (1.62)	82.64 (1.04)	90.55 (1.68)	80.65 (2.42)	89.13 (1.23)	91.95 (0.69)	93.27 (0.73)	**96.71** (1.06)
Time (s)	1.24	10.59	116.54	17.24	2536.81	943.62	1809.25	1021.10	2757.34	3440.03	2912.57

TABLE VIII

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA, AA, AND κ ON THE HOUSTON2013 DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class No.	KNN	SVM	AL-MLR	CCFs	SSAL	2D-CNNs	3D-CNNs	v-CNNs	AL-CNNs	AM-CNNs	SA-CNNs
1	84.92	90.33	84.34	88.77	90.66	85.08	89.02	85.16	86.90	84.78	**97.31**
2	92.05	96.64	**99.92**	95.98	94.65	96.15	98.03	97.46	99.84	96.09	93.02
3	98.96	99.41	99.70	**100.00**	99.85	93.76	92.57	94.35	99.70	98.96	99.70
4	99.34	99.26	99.67	99.42	99.75	96.21	97.94	97.53	98.94	97.71	**100.00**
5	93.45	97.43	97.18	98.92	95.50	**100.00**	98.92	99.92	98.85	**100.00**	99.92
6	92.03	83.06	91.36	96.68	**99.67**	87.71	87.38	78.07	84.77	81.46	81.73
7	77.71	90.60	93.35	91.73	91.59	85.49	86.22	87.20	93.38	92.48	**95.61**
8	60.82	66.58	34.32	59.34	63.32	77.45	80.58	61.56	88.60	85.61	**89.25**
9	69.93	70.18	81.47	82.04	79.63	81.38	88.96	80.89	89.74	**92.64**	90.47
10	56.46	81.65	70.56	90.49	95.86	91.33	74.56	71.23	88.46	93.51	**96.30**
11	77.36	83.38	86.80	94.49	94.65	94.99	86.47	93.07	90.17	**97.84**	94.68
12	58.33	71.67	76.75	76.25	74.36	87.75	81.67	**95.83**	90.73	85.46	89.30
13	22.99	40.63	66.52	43.08	91.98	72.10	58.04	84.38	92.47	**94.75**	94.14
14	95.56	99.75	**100.00**	99.51	**100.00**	89.38	92.84	88.15	97.78	95.07	93.61
15	97.31	98.26	**100.00**	98.58	**100.00**	**100.00**	96.37	99.05	99.53	99.69	99.69
OA (%)	78.13 (0.93)	85.08 (1.31)	84.20 (1.74)	87.92 (0.91)	89.76 (1.02)	89.63 (1.58)	87.99 (1.66)	87.61 (1.43)	93.20 (1.25)	93.13 (1.30)	**94.75** (1.15)
AA (%)	78.48 (0.94)	84.59 (1.39)	85.46 (1.21)	87.69 (1.01)	91.43 (1.10)	89.25 (1.83)	87.30 (1.57)	87.59 (1.65)	93.32 (1.08)	93.07 (1.11)	**94.31** (1.10)
κ	76.35 (1.01)	83.85 (1.42)	82.94 (1.44)	86.93 (0.99)	89.76 (1.07)	88.78 (1.21)	87.00 (1.00)	86.60 (1.43)	92.65 (1.18)	92.57 (1.12)	**94.32** (0.89)
Time (s)	1.15	46.42	408.56	15.20	4121.03	988.81	1974.34	1293.96	3560.59	4528.80	3885.19

the confidence of the classifier in predicting actively selected reliable samples using their pseudolabels. On the other hand, for the classes with a high degree of uncertainties, an interactive AL process with limited cost plays a vital role in keeping a satisfying class balance. This simultaneous consideration ensures the superiority of our methods with regard to both the OA and AA indexes.

3) Houston2013 Dataset: For the final dataset, we empirically tuned 10 000 superpixel numbers, 150 budgets, and the log AL schedule for our SA-CNNs. Table VIII gives per-class quantitative results, and Fig. 7 demonstrates visual results of all the methods compared.

The large scene covered by Houston2013 makes the issue of class imbalance in this dataset much better than the previous two. Apart from the baseline KNN, the results of SVM, AL-MLR, CCFs, 3D-CNNs, and v-CNNs are close to each other. Once again 2D-CNN shows slight advantages owing to proper dimensionality reduction on spectra. Dramatically, there arises an insignificant performance decrease from AL-CNNs to AM-CNNs with less than 0.3%, which means that simple SSL strategy cannot always guarantee a performance improvement. The training of networks can mislead into an undesirable direction due to false pseudolabels. This might also be explained by the errors caused by oversmoothing the disjoint zones consisting of different land cover categories, for example, misclassified *Stressed Grass* area beneath the cross of *Railway* and *Highway* on the right side. This issue is addressed in the result of SA-CNNs. In particular, for the class of *Health Grass* that is easily confused with other classes of grass (*Stressed Grass*, *Synthetic Grass*, and even *Tree*), our method is able to win

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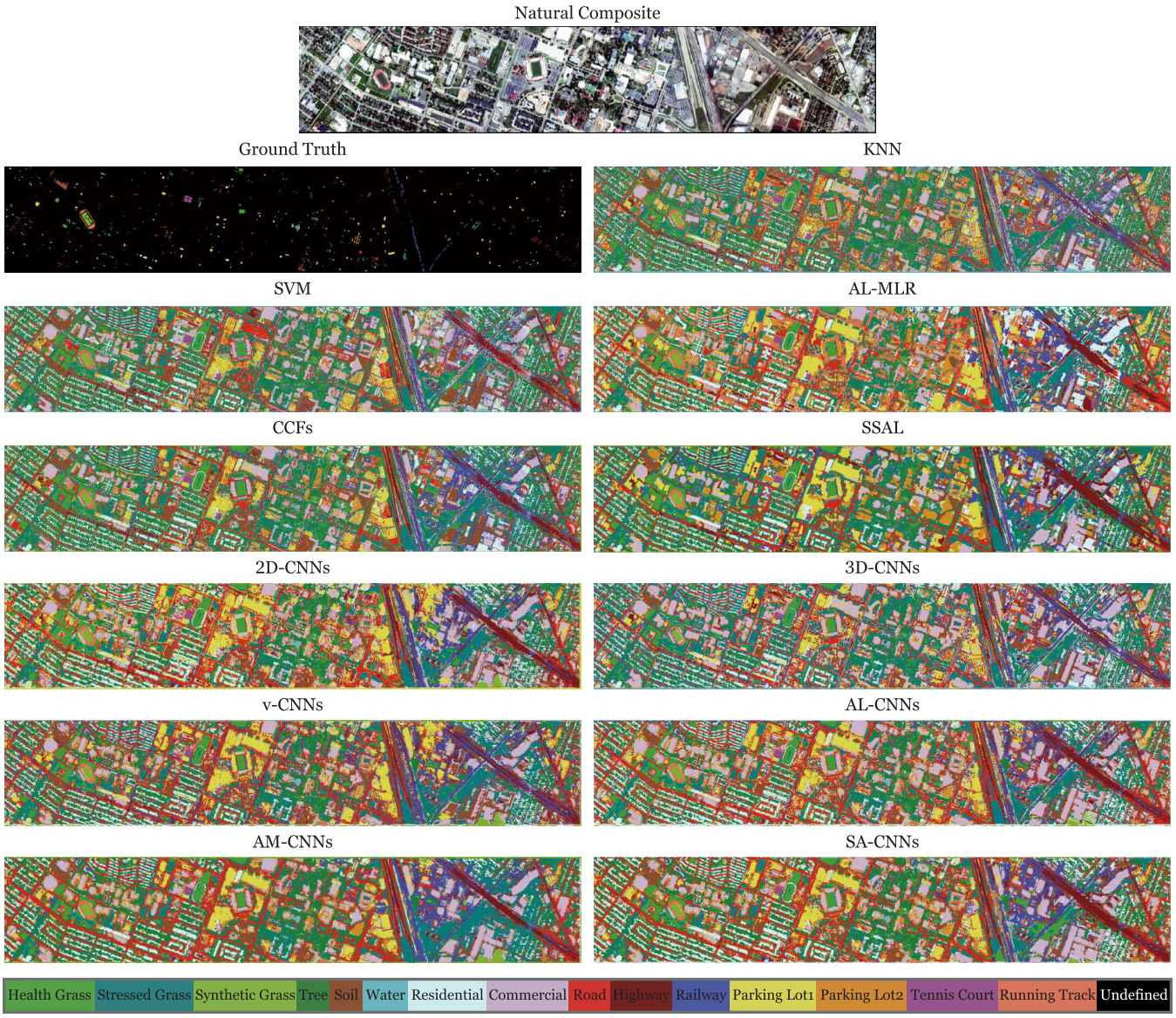


Fig. 7. Illustration of natural composite image, label ground truth, and classification maps obtained by the compared methods on the Houston2013 dataset.

around 10% performance improvement with regard to other approaches.

V. Conclusion

In this article, we have introduced a novel CNN-based method called SA-CNNs for HSI classification. Based on the statistical observation that superpixel segmentation can generate a large proportion of superpixels with highly consistent labels, we propose to simultaneously select labeled and unlabeled samples within different superpixels in a semi-active manner. To differ from the existing AL and SSL-based methods, through an iterative training of the deep classifier, the complementary limited supervision and reliable pseudosupervision can be jointly used with mutual promotions under our framework. Moreover, three rational and effective schedules are devised to fully exploit the potential of AL in dealing with the HS data with various distributions and train–test set split. Extensive ablation and comparison experiments conducted on the benchmark HSI datasets verify that the proposed SA-CNN is the best overall performer, leading other competitors by obvious margins. This indicates the practicality and superiority of the proposed framework in tackling remotely sensed HS scenes with limited labeling availability but plenty of unlabeled data.

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REFERENCES

[1] D. Hong *et al.*, "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.

[2] T. Lu, S. Li, L. Fang, Y. Ma, and J. A. Benediktsson, "Spectral–spatial adaptive sparse representation for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 373–385, Jan. 2016.

[3] X. Fu, W. Wang, Y. Huang, X. Ding, and J. Paisley, "Deep multiscale detail networks for multiband spectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 2090–2104, May 2021.

[4] B. Zhang, S. Li, X. Jia, L. Gao, and M. Peng, "Adaptive Markov random field approach for classification of hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 5, pp. 973–977, Sep. 2011.

[5] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[6] D. Hong, N. Yokoya, N. Ge, J. Chanussot, and X. X. Zhu, "Learnable manifold alignment (LeMA): A semi-supervised cross-modality learning framework for land cover and land use classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 147, pp. 193–205, Jan. 2019.

[7] M. H. R. Sales, S. de Bruin, C. Souza, and M. Herold, "Land use and land cover area estimates from class membership probability of a random forest classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[8] C.-I. Chang, "An information-theoretic approach to spectral variability, similarity, and discrimination for hyperspectral image analysis," *IEEE Trans. Inf. Theory*, vol. 46, no. 5, pp. 1927–1932, Aug. 2000.

[9] B. Zhang, X. Sun, L. Gao, and L. Yang, "Endmember extraction of hyperspectral remote sensing images based on the ant colony optimization (ACO) algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 7, pp. 2635–2646, Jul. 2011.

[10] J. Yao, D. Hong, L. Xu, D. Meng, J. Chanussot, and Z. Xu, "Sparsity-enhanced convolutional decomposition: A novel tensor-based paradigm for blind hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3069845.

[11] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3064958.

[12] B. Zhang, W. Yang, L. Gao, and D. Chen, "Real-time target detection in hyperspectral images based on spatial–spectral information extraction," *EURASIP J. Adv. Signal Process.*, vol. 2012, no. 1, pp. 1–15, Dec. 2012.

[13] X. Wu, D. Hong, J. Tian, J. Chanussot, W. Li, and R. Tao, "ORSIm detector: A novel object detection framework in optical remote sensing imagery using spatial-frequency channel features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5146–5158, Jul. 2019.

[14] X. Wu, D. Hong, J. Chanussot, Y. Xu, R. Tao, and Y. Wang, "Fourier-based rotation-invariant feature boosting: An efficient framework for geospatial object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 302–306, Feb. 2020.

[15] Y. Gu, T. Wang, X. Jin, and G. Gao, "Detection of event of interest for satellite video understanding," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7860–7871, Nov. 2020.

[16] J. Yao, D. Hong, J. Chanussot, D. Meng, X. Zhu, and Z. Xu, "Cross-attention in coupled unmixing nets for unsupervised hyperspectral super-resolution," in *Proc. ECCV*. Cham, Switzerland: Springer, 2020, pp. 208–224.

[17] L. Gao, D. Hong, J. Yao, B. Zhang, P. Gamba, and J. Chanussot, "Spectral superresolution of multispectral imagery with joint sparse and low-rank learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2269–2280, Mar. 2021.

[18] X. Cao, X. Fu, D. Hong, Z. Xu, and D. Meng, "PanCSC-Net: A model-driven deep unfolding method for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, doi: 10.1109/TGRS.2021.3115501.

[19] T. Li and Y. Gu, "Progressive spatial–spectral joint network for hyperspectral image reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[20] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[21] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.

[22] A. J. Izenman, "Linear discriminant analysis," in *Manifold Learning Theory and Applications*. Cham, Switzerland: Springer, 2013, pp. 237–280.

[23] Y. Ma and Y. Fu, *Manifold Learning Theory and Applications*. Boca Raton, FL, USA: CRC Press, 2011.

[24] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "CoSpace: Common subspace learning from hyperspectral-multispectral correspondences," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4349–4359, Jul. 2019.

[25] H. L. Yang and M. M. Crawford, "Spectral and spatial proximity-based manifold alignment for multitemporal hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 51–64, Jan. 2016.

[26] Z. He, L. Liu, R. Deng, and Y. Shen, "Low-rank group inspired dictionary learning for hyperspectral image classification," *Signal Process.*, vol. 120, pp. 209–221, Mar. 2016.

[27] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[28] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Jul. 2016.

[29] B. Rasti *et al.*, "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[30] X. Cao, X. Fu, C. Xu, and D. Meng, "Deep spatial–spectral global reasoning network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3069241.

[31] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[32] D. Hong *et al.*, "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, doi: 10.1109/TGRS.2021.3130716.

[33] D. Hong *et al.*, "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Aug. 2020.

[34] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020.

[35] D. Hong, J. Yao, D. Meng, Z. Xu, and J. Chanussot, "Multimodal GANs: Toward crossmodal hyperspectral–multispectral image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5103–5113, Jun. 2021.

[36] A. Samat, J. Li, C. Lin, S. Liu, and E. Li, "Edge gradient-based active learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 9, pp. 1588–1592, Sep. 2020.

[37] C. Persello and L. Bruzzone, "Active and semisupervised learning for the classification of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6937–6956, Nov. 2014.

[38] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[39] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Hyperspectral image segmentation using a new Bayesian approach with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3947–3960, Oct. 2011.

[40] W. Di and M. M. Crawford, "View generation for multiview maximum disagreement based active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 5, pp. 1942–1954, May 2012.

[41] Q. Shi, B. Du, and L. Zhang, "Spatial coherence-based batch-mode active learning for remote sensing image classification," *IEEE Trans. Image Process.*, vol. 24, no. 7, pp. 2037–2050, Jul. 2015.

[42] P. Liu, H. Zhang, and K. B. Eom, "Active deep learning for classification of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 2, pp. 712–724, Feb. 2017.

[43] J. M. Haut, M. E. Paoletti, J. Plaza, J. Li, and A. Plaza, "Active learning with convolutional neural networks for hyperspectral image classification using a new Bayesian approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6440–6461, Nov. 2018.

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[44] X. Cao, J. Yao, Z. Xu, and D. Meng, "Hyperspectral image classification with convolutional neural network and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4604–4616, Jul. 2020.
[45] K. Sohn *et al.*, "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 596–608.
[46] I. Dópido, J. Li, P. R. Marpu, A. Plaza, J. M. B. Dias, and J. A. Benediktsson, "Semisupervised self-learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 7, pp. 4032–4044, Jul. 2013.
[47] H. Wu and S. Prasad, "Semi-supervised deep learning using pseudo labels for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1259–1270, Mar. 2018.
[48] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Learning to propagate labels on graphs: An iterative multitask regression framework for semi-supervised hyperspectral dimensionality reduction," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 35–49, Dec. 2019.
[49] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Joint and progressive subspace analysis (JPSA) with spatial–spectral manifold alignment for semisupervised hyperspectral dimensionality reduction," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3602–3615, Jul. 2021.
[50] J. Munoz-Mari, D. Tuia, and G. Camps-Valls, "Semisupervised classification of remote sensing images with active queries," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3751–3763, Oct. 2012.
[51] Z. Zhang, E. Pasolli, M. M. Crawford, and J. C. Tilton, "An active learning framework for hyperspectral image classification using hierarchical segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 640–654, Feb. 2016.
[52] C. Liu, J. Li, and L. He, "Superpixel-based semisupervised active learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 1, pp. 357–370, Jan. 2019.
[53] S. Li, T. Lu, L. Fang, X. Jia, and J. A. Benediktsson, "Probabilistic fusion of pixel-level and superpixel-level hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7416–7430, Dec. 2016.
[54] T. Lu, S. Li, L. Fang, X. Jia, and J. A. Benediktsson, "From subpixel to superpixel: A novel fusion framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4398–4411, Aug. 2017.
[55] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2011.
[56] V. Jampani, D. Sun, M.-Y. Liu, M.-H. Yang, and J. Kautz, "Superpixel sampling networks," in *Proc. ECCV*, Sep. 2018, pp. 352–368.
[57] A. J. Joshi, F. Porikli, and N. Papanikolopoulos, "Multi-class active learning for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 2372–2379.
[58] T. Rainforth and F. Wood, "Canonical correlation forests," 2015, *arXiv:1507.05444*.
[59] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.
[60] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.



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